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Calculus 1 · Week 2

Functions and graphs

A function gives each input exactly one output; y = f(x).

- Evaluate by replacing every x with the input in parentheses: f(a + 1) is not f(a) + 1.
- Vertical line test: a curve is a function of x if no vertical line meets it twice.

> **Think of it like this.** A function is a vending machine: you press one button (the input) and exactly one snack drops (the output). Two buttons may give the same snack, but a machine that sometimes gives chips and sometimes a cookie for the same button is broken, and so is a "function" with two outputs for one input.

Domain: remove zeros of denominators, negatives under even roots, and non-positive log arguments.

- Several restrictions: solve each one and intersect the answers.
- Range: read the outputs from the graph, or solve $y = f(x)$ for x.

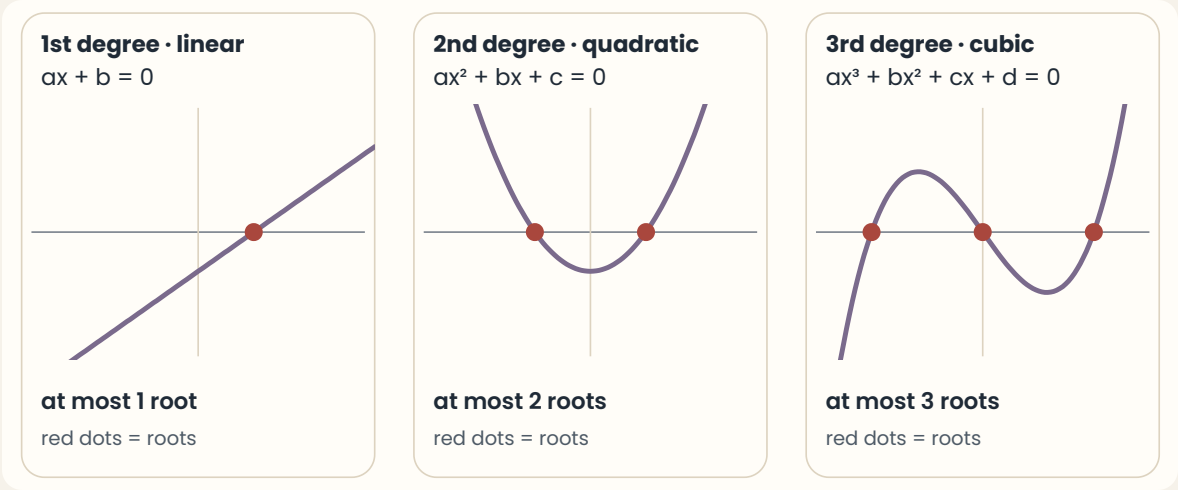
> **Think of it like this.** The domain is the coin slot of a machine: it only takes coins it recognises. The range is the shelf of everything the machine can actually give you, which is often smaller than "every number".

Increasing: going up from left to right; decreasing: going down. Intervals are written with x values.

- Even: f(-x) = f(x), mirror in the y-axis. Odd: f(-x) = -f(x), half-turn about the origin.
- Domain and range are the shadows of the graph on the two axes.

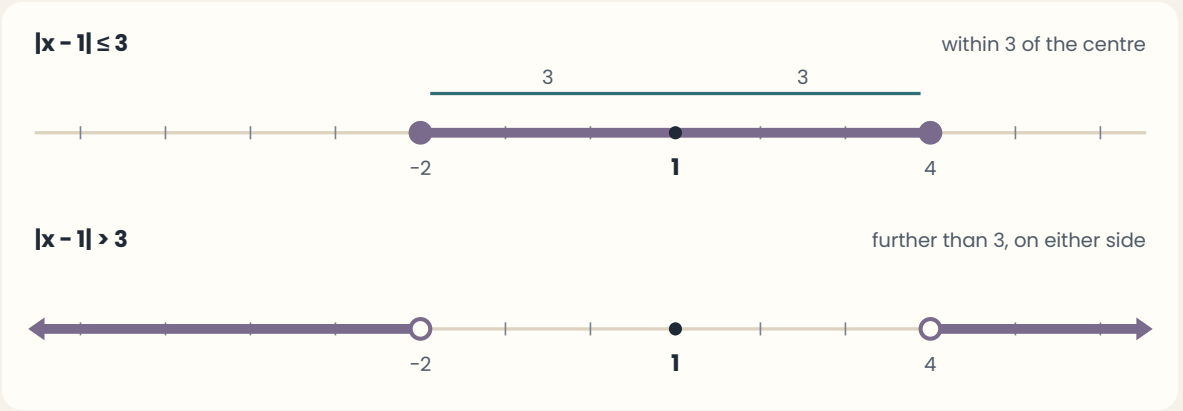
> **Think of it like this.** Read a graph like the elevation profile of a hiking trail, walking from left to right: uphill stretches are where the function increases, downhill where it decreases, summits are local maxima, and valleys are local minima.

Linear: slope $m = (y_2 - y_1)/(x_2 - x_1)$, intercept $b = f(0)$; $m > 0$ rises, $m < 0$ falls.



> **Think of it like this.** Function families are like vehicle types: once you know you are looking at a bicycle, a bus, or a boat, you already know roughly how it moves. Recognising the family from the formula tells you the shape before you plot a single point.

Evaluate a piecewise function by choosing the one piece whose condition holds.



> **Think of it like this.** A piecewise function is like a phone plan: one price while you stay under your data allowance, another rule after you pass it. To know what you pay, first check which band you are in, then apply only that band's rule.

f(x) + k: up k. f(x − h): right h. f(x + h): left h.

- a·f(x): vertical stretch; −f(x): flip over the x-axis; f(−x): flip over the y-axis.
- f ± g, fg, f/g live on the intersection of the domains; f/g also needs g(x) ≠ 0.

> **Think of it like this.** Transforming a graph is like moving a sticker on a window: you can slide it, stretch it, or flip it over, and it is still the same picture. You only need to track where one landmark point (like the vertex) goes.

$(f \circ g)(x) = f(g(x))$: g first. Usually $f \circ g \neq g \circ f$.

- One-to-one: horizontal line test. Only one-to-one functions have inverses.
- Inverse: $y = f(x)$, solve for x, rename; the graph mirrors f across $y = x$.

> **Think of it like this.** Composition is an assembly line: the part leaves the first machine and goes straight into the second. The inverse is the undo button: whatever f did, f^{-1} takes back, like rewinding a video to the exact frame where you started.

180° = π rad; multiply degrees by π/180 to get radians.

- Unit circle point: $(\cos\theta, \sin\theta)$; $\sin^2\theta + \cos^2\theta = 1$; $\tan = \sin/\cos$.
- $y = A\sin(Bx) + D$: amplitude $|A|$, period $2\pi/|B|$, range $[D - |A|, D + |A|]$.

> **Think of it like this.** Picture a seat on a Ferris wheel. Its height above the ground rises and falls in a smooth wave: that is a sine curve. One full turn of the wheel is one period, and the radius of the wheel is the amplitude.

a^x ($a > 0$, $a \neq 1$): domain $\mathbb{R}$, range $(0, \infty)$, through $(0, 1)$; grows if $a > 1$, decays if $0 < a < 1$.

- $\log_a b = x \Leftrightarrow a^x = b$. Logs turn products into sums and powers into factors.
- Solve: same base → equal exponents; otherwise take logs. Always check log arguments > 0.

> **Think of it like this.** An exponential is a rumour that doubles every hour: slow at first, then explosive. A logarithm asks the opposite question: "how many doublings did it take to reach this many people?"

Step by step

Build $y = -2(x + 1)^2 + 4$

1. Start with $y = x^2$, vertex $(0, 0)$
2. Move left 1: $y = (x + 1)^2$
3. Stretch vertically by 2: $y = 2(x + 1)^2$
4. Reflect across the x-axis: $y = -2(x + 1)^2$
5. Move up 4: vertex $(-1, 4)$

Check yourself

1. Which rule is NOT a function?

A) Each person → their date of birth B) Each date → the people born that day C) Each product → its price today

2. What is the range of $f(x) = 5 - x^2$?

A) $(-\infty, 5]$ B) $[5, \infty)$ C) $\mathbb{R}$

3. A café's daily profit is $P(t) = -2t^2 + 24t - 40$ (t = hours after opening). On which interval is profit rising?

A) $[0, 6]$ B) $[6, 12]$ C) $[0, 32]$

Answers are in the speaker notes.

Practise next

Labs

- Evaluate, find domains, and read graphs
- Families, pieces, and transformations
- Composition, inverses, and waves
- Exponentials and logarithms at work

On the site

- Graph Lab